PRD: Verified Fee Reconciliation

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1. Background and problem

Students pay admission and tuition fees by UPI, bank transfer, or cash collected by authorised staff. Payments often do not carry the student's registered mobile number or email:

- A parent pays from their own account, or pays a later instalment.
- A student pays from a friend's UPI ID.
- One transfer covers two students.
- A student pays in several parts.
- Money arrives before the student's record is complete.

Operations matches payments to students by hand. Current impact: incorrect fee balances, delayed receipts, repeated follow-ups with students, unidentified credits, and duplicate entries. Baselines for each are to be measured in week 1 (section 2.3).

Founder mandate: solve this with an application integrated with the existing student-management system (SMS).

The APM draft is not approved. It treats screenshots as proof of payment, matches on student name, and lets counsellors edit payments and mark cash as paid. It also has no rules for duplicates, splits, reversals or refunds. Full review: Appendix A.

2. Goals, non-goals and success metrics

2.1 Goals

1. Every bank-confirmed credit ends up either allocated to the correct student and fee component, or in a suspense queue with an owner and a due date.
2. A student's balance and receipts reflect only money confirmed by the bank, payment-gateway or UPI feed.
3. Payments that can be identified with certainty are allocated without manual work. All others are routed to the right reviewer with the evidence needed to decide.
4. Every allocation, correction and approval can be audited.

2.2 Non-goals

- Removing manual review entirely. Some payments are ambiguous; they need a person.
- Changing fee structures, scholarships or discounts.

- Building an online checkout, or replacing the SMS or accounting system.

2.3 Success metrics

Metric	Definition	Target	Source
Unverified credits	Ledger credits with no linked bank / gateway transaction	0	Ledger audit report
Duplicate receipts	Receipts issued more than once for the same transaction	0	Receipt register
Auto-allocation rate	Credits auto-allocated ÷ total bank credits, monthly	≥ 60% (phase 1)	Matching report
Suspense resolution	Suspense items resolved within 2 working days ÷ items created	≥ 90%	Ageing report
Time to receipt	Bank import to receipt issued, auto-allocated credits, 90th percentile	≤ 1 hour	Event timestamps
Mismatch tickets	Student mismatch tickets per 100 payments	−50% vs baseline	Ticket log
Daily tie-out	Bank credits − (ledger postings + suspense)	₹0 unexplained	Tie-out report

Baselines come from the previous term's manual records in week 1 of build. The auto-allocation target will be reviewed once we know what share of credits carry a student ID.

3. Users and roles

Role	Responsibility	Primary actions in the system
Student	Pays fees; tracks balance	View ledger; submit payment details; raise mismatch; download receipts. Parents act through the student's login
Counsellor	Supports students; collects cash at centres	Create cash collection notes; submit payment details on behalf of assigned students
Ops executive	Resolves unmatched payments	Work suspense queue; contact students and payers; propose allocations
Ops lead	Queue health, SLA, automation quality	Approve allocations backed by ID-grade evidence; sample auto-allocations; handle escalations
Finance	Books and bank reconciliation	Verify cash; approve high-risk allocations, reallocations and refunds; review reversals; daily tie-out
Finance head	Financial control	Approve refunds and changes to rules or parameters
Admin	System configuration	Manage users, roles and parameters (with approval); read audit log

4. Scope, assumptions and dependencies

4.1 Scope

Release	Includes
Phase 1	Bank / UPI / gateway feed intake · duplicate detection · matching and allocation rules · suspense queue · student claims · cash collection notes · reversals and refunds · fee ledger by component · receipts and credit notes · approvals · SMS balance sync · notifications · audit log · reports (ageing, tie-out, matching, receipts)
Phase 2	Per-student virtual account or payment link (pending founder decision) · screenshot OCR to pre-fill claims · parents register payer accounts themselves · accounting-system export · auto-allocation of exact-UTR claims if phase 1 sampling error rate is < 0.5%

4.2 Assumptions

ID	Assumption	Validated by
AS1	A bank, gateway or UPI feed with transaction ID / UTR, amount, value date, payer name and narration is available at least daily	Finance + Engineering
AS2	The SMS holds student ID, status, registered and guardian contacts, fee components and due dates, and can share them by API or export	Engineering
AS3	Student IDs follow a fixed format (S + 3 digits in the sample data)	Operations
AS4	Cash collected at centres is deposited into the college's bank account and appears in the feed with a deposit reference	Finance
AS5	Students can be asked to put their student ID in the payment reference	Operations

4.3 Dependencies

Dependency	Owner	Risk if late
Bank / gateway feed access	Finance	No source of truth. Fallback: daily statement file upload
SMS API for student data, fee components, balance write-back	Engineering / SMS vendor	Balances diverge between systems
Opening balances for current students	Finance	Go-live blocked
Notification channels and templates (SMS, WhatsApp, email)	APM + Operations	Students not informed; more tickets
Finance sign-off on rules and parameters (section 7)	PM	Build of matching and approvals blocked

5. Solution overview

5.1 Product decisions

#	Decision	Rationale
D1	Only the bank, gateway or UPI feed is proof of payment	Screenshots can be edited or show failed payments. Counsellor reports are not deposits. Both stay as matching evidence
D2	Receiving money and allocating it are separate steps	Money can be confirmed while its owner is still unknown (P2, P9). Suspense holds it without touching any balance
D3	Auto-allocate only on an ID-grade identifier: student ID, virtual account, or verified payer account	A name or contact match cannot tell apart students with the same surname, or a parent paying for a sibling

#	Decision	Rationale
D4	The ledger is append-only and balances are calculated from it	Corrections stay visible to students and auditors; no balance can be typed in
D5	Maker-checker on every manual allocation, cash verification, reallocation and refund	Separates the person who records from the person who approves
D6	This application owns payments, allocations and receipts; the SMS receives balances	One record per fact; balance fields in the SMS become read-only. To be confirmed in the engineering walkthrough (Q2)

5.2 Workflow

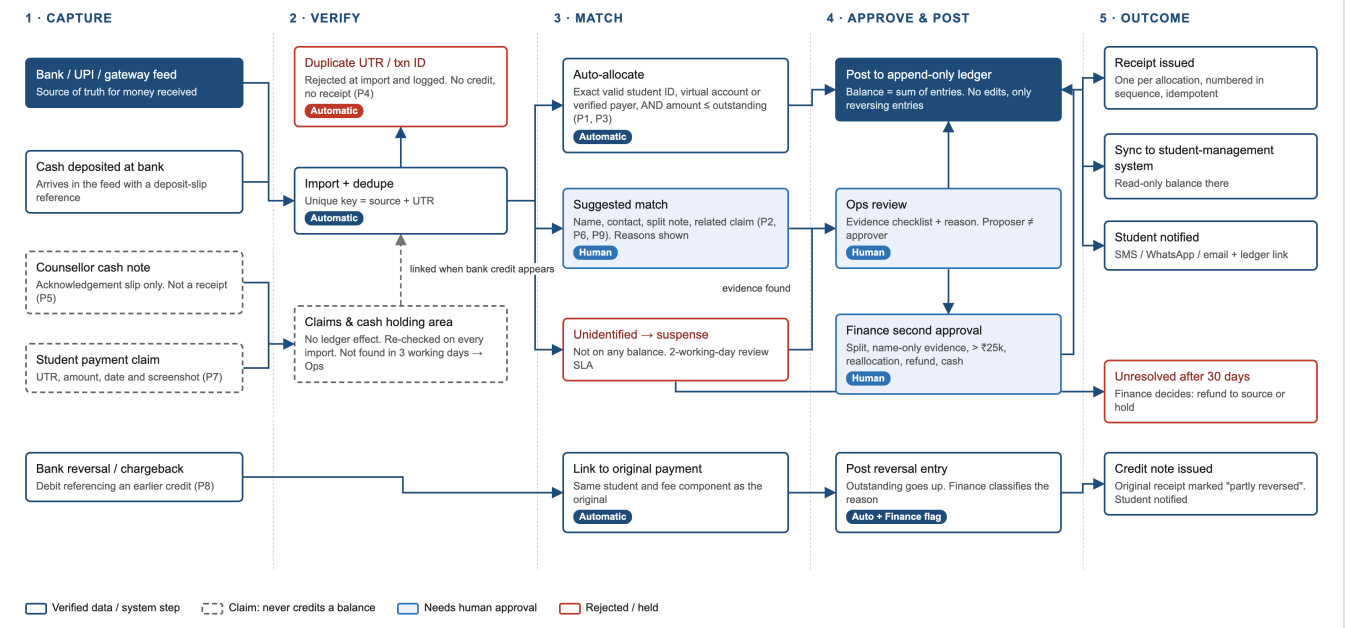


Figure 1. Payment workflow. Dashed: claims (no ledger effect). Blue: human approval. Red: rejected or held.

5.3 Statuses

Payment status	Meaning	Shown to student as
Duplicate	Same source and UTR already imported. No further processing	Not shown
Unidentified	Bank-confirmed; no ID-grade identifier; no suggestion	Not shown (not yet linked to a student)
Suggested	Bank-confirmed; one or more candidate students with reasons; awaiting review	Not shown
Pending approval	Allocation proposed; awaiting approver	"Payment received, being processed"
Allocated	Posted to the ledger; receipt issued	"Paid", with receipt
Partly reversed / Reversed	Bank debit linked; reversal posted; credit note issued	"Reversed by bank", with credit note
Refunded	Refund approved and confirmed by bank debit	"Refunded"

Claim / cash note status	Meaning	Shown to student as
Awaiting bank	Submitted; no matching bank credit yet; rechecked on every import	"Awaiting bank confirmation"

Claim / cash note status	Meaning	Shown to student as
Linked	Matched to a bank credit; the allocation follows payment rules	Status of the linked payment
Not found	No matching credit after the recheck window; with Ops	"We could not find this payment. Ops will contact you"
Cash: Reported → Verified	Counsellor note created → Finance matched it to a deposit credit	"Cash acknowledged, pending verification" → "Paid"

6. Functional requirements

Priority: **High** = phase 1 release blocker. **Medium** = phase 1; may follow within 2 weeks of launch. **Low** = moves to the next release if capacity is short.

ID	Requirement	Priority
Payment intake		
FR-01	Import credits and debits from each configured bank, gateway and UPI source at the frequency agreed with engineering. Statement-file upload by Finance is available as a fallback	**High**
FR-02	Reject a transaction whose source and UTR / transaction ID already exist. Log it in the import report with a reference to the original	**High**
FR-03	Flag, but do not reject, a transaction with a different UTR but the same amount, value date and payer as an existing one ("possible duplicate")	**Medium**
FR-04	Quarantine malformed rows with a reason, without blocking the rest of the import	**Medium**
Matching and allocation		
FR-05	Read student IDs from the payment reference / narration	**High**
FR-06	Auto-allocate a credit when all of these hold: exactly one ID-grade identifier resolving to one active student; amount ≤ that student's outstanding; no open dispute (BR3)	**High**
FR-07	Otherwise, generate suggestions with the reasons and a confidence level (High / Medium / Low, section 7.2), and place the credit in the suspense queue	**High**
FR-08	When the reference names several student IDs, pre-fill a split proposal using any amounts in the reference	**Medium**
FR-09	Apply allocated money to fee components in due-date order unless Finance overrides with a reason (BR5)	**High**
FR-10	Hold any amount above total outstanding as a credit balance and route it to Finance (BR6)	**High**
FR-11	Re-run matching for suspense credits when a student record is created or updated, or a claim is submitted	**Medium**
FR-12	Save a payer account as verified for a student after a confirmed allocation, with the student's confirmation or a recorded call / email	**Low**
Student claims		
FR-13	Student submits a claim: UTR (required), amount, payment date, payer name, screenshot (JPG / PNG / PDF, ≤ 5 MB)	**High**
FR-14	Link a claim to an unallocated credit with the same UTR and amount, and propose allocation to the claiming student (approver: Ops lead)	**High**
FR-15	Recheck unlinked claims on every import for 3 working days, then set them to Not found and assign to Ops	**Medium**
FR-16	Flag a conflict when a UTR is claimed by more than one student, or for a credit that is already allocated	**Medium**

ID	Requirement	Priority
Cash		
FR-17	Counsellor creates a cash collection note: student ID, amount, collection date, acknowledgement-slip number. The student receives an acknowledgement, not a receipt	**High**
FR-18	Finance links a cash note to a deposit credit and verifies it. The allocation then follows standard rules	**High**
FR-19	Escalate notes not verified within 2 working days to the Finance head, and block that counsellor's new notes until cleared	**Medium**
Reversals and refunds		
FR-20	Link a bank debit that references an earlier credit; post a reversal to the same student and component; issue a credit note; flag Finance	**High**
FR-21	Route debits with no reference, and reversals larger than the amount allocated, to Finance without posting	**High**
FR-22	Refund flow: Finance proposes, Finance head approves, posted when the bank debit is imported	**Medium**
Ledger, receipts and corrections		
FR-23	Keep an append-only ledger per student and fee component. Balances are calculated from entries; no user can edit a balance or an entry	**High**
FR-24	Issue one receipt per posted allocation with a sequential number. Re-download returns the same document	**High**
FR-25	Correct an allocation by posting a reversal entry plus a new allocation; cancel the old receipt, issue a new one, and notify the affected students	**High**
FR-26	Send posted balances to the SMS; retry on failure and show "sync pending" on the student record	**High**
Notifications, reports and audit		
FR-27	Notify students on: claim received, allocation posted (with receipt link), reversal, refund, request for information. Messages never contain another person's name, account or contact	**Medium**
FR-28	Reports: suspense ageing; daily bank-to-ledger tie-out; auto-allocation sample list; matching rate; receipt register; claims not found	**Medium**
FR-29	Record every system and user action with actor, timestamp, before / after values and reason. The record cannot be edited and can be exported	**High**
FR-30	Student mismatch ticket linked to a ledger line or claim, with a 2-working-day response SLA	**Low**

7. Business rules and parameters

7.1 Rules

ID	Rule
BR1	A balance changes only on a ledger entry linked to a bank, gateway or UPI transaction, or to a fee charge from the SMS
BR2	Screenshots, counsellor reports, payer names and contact numbers are matching evidence only
BR3	Auto-allocation needs all of these: bank-confirmed; one ID-grade identifier resolving to one active student; amount ≤ outstanding; no open dispute or claim conflict
BR4	Never auto-allocated: name / contact / amount matches, splits across students, overpayments, cash, refunds, reallocations, inactive students
BR5	Fee component order: earliest due date first (admission fee before tuition instalments). Finance may override with a recorded reason

ID	Rule
BR6	An overpayment is held as a credit balance. Finance chooses whether to apply it to the next instalment or refund it
BR7	A split needs a valid student ID on every line, and the lines must total the credit exactly. One receipt per student
BR8	The person who proposes an allocation cannot approve it
BR9	A credit for an unregistered student stays in suspense ("pending registration") and is re-matched when the record is created
BR10	Suspense credits unresolved at the suspense limit go to Finance for a refund-to-source decision
BR11	Payer account numbers are masked to the last 4 digits for every role except Finance

7.2 Suggestion confidence

Level	Evidence	Approver
High	Claim UTR and amount match the credit exactly · payer account verified for this student (when the payer is linked to more than one student)	Ops lead
Medium	Payer name equals the student's full name · payer contact equals the registered or guardian contact	Finance
Low	Surname or partial name match · amount equals an instalment due · a claim with the same amount and date but no matching UTR	Finance

7.3 Parameters

Defaults below. Admin changes them with Finance-head approval; every change is logged.

Parameter	Default	Effect
Manual allocation needing Finance approval	> ₹25,000	Above this, an Ops lead approval is not enough
Auto-allocation sample for Ops lead review	5% daily	Quality check on automation
Claim recheck window	3 working days	Then claim is set to Not found
Suspense review SLA	2 working days	Escalation to Ops lead
Suspense limit	30 days	Finance refund-to-source decision
Cash deposit window	2 working days	Escalation to Finance head
Mismatch ticket SLA	2 working days	Escalation to Ops lead

8. User journeys

Role	Journey
Student	1. Pays with student ID in the reference (shown on the fee page and invoice). 2. Optionally submits UTR + screenshot; the ledger shows "Awaiting bank confirmation" with no balance change. 3. When allocated, receives a notification with a receipt link. 4. Views components, payments, pending claims, reversals and outstanding amounts. 5. Raises a mismatch on any ledger line.
Ops	1. Opens the suspense queue, sorted by amount × age. 2. Opens a credit: bank record, candidates with reasons, missing evidence. 3. Requests information through templates. 4. Attaches evidence and a reason, and proposes allocation. 5. The system routes it to the approver per section 11. Unresolved items escalate at the SLA and at the suspense limit.
Finance	1. Daily tie-out: bank credits = ledger postings + suspense. 2. Links cash notes to deposit credits and verifies them. 3. Reviews reversal reasons. 4. Approves high-risk allocations, reallocations and refunds. 5. Decides on overpayments and suspense items past the limit.

Role	Journey
Admin	1. Manages users and roles. 2. Proposes parameter changes (Finance head approves). 3. Exports audit records. No access to create or modify ledger entries.

Case	Behaviour	Who acts
Automatic match	Posted, receipt issued, student notified, included in daily sample	System; Ops lead samples
Suspected match	Suggested status; candidates with reasons; no balance change	Ops proposes; approver by confidence level
Missing details	Unidentified; re-matched on student creation or claim; escalated at SLA	Ops
Multiple payments	Each credit allocated independently; due-date order; any excess goes to credit balance	System; Finance for excess
Payment for several students	Split proposal pre-filled from the reference; must total exactly	Ops proposes; Finance approves
Reversal	Linked to original; reversal entry; credit note; student notified	System; Finance reviews reason

9. Sample data walkthrough

Expected handling of the nine sample records. This table is also test fixture T-SAMPLE (section 15.2).

Rec	Amount	Rule path	Result	Effect on outstanding / documents	Cannot yet conclude	Needed to allocate
P1	₹10,000	Bank-confirmed; one valid ID (S101); amount ≤ outstanding	Allocated (auto)	S101 admission −₹10,000 · receipt	Nothing	Nothing. Sender account offered for linking to S101 (FR-12)
P2	₹20,000	Bank-confirmed; no ID; surname match S102; related claim P7	Suggested (Low)	None	Owner. "R Mehta" may not be S102 or a guardian	Claim UTR = bank UTR, or payer confirmed as S102's guardian. Finance approves
P3	₹15,000	Bank-confirmed; one valid ID (S103)	Allocated (auto)	S103 admission −₹15,000 · receipt	Nothing	Nothing
P4	₹10,000	Same UTR as P1 (FR-02)	Duplicate	None · no receipt	Nothing	Nothing. Logged in import report
P5	₹10,000	Cash note; no deposit credit	Cash: Reported	None · acknowledgement only	Whether the cash reached the college; which student and component, if not on the note	Deposit credit linked by Finance; acknowledgement-slip number; student ID
P6	₹30,000	Bank-confirmed; two IDs with amounts; total matches	Pending approval (split)	On approval: S101 −₹15,000, S104 −₹15,000 · 2 receipts	Whether S101 is overpaid (₹25,000 paid in total)	Finance approval; overpayment check per student

Rec	Amount	Rule path	Result	Effect on outstanding / documents	Cannot yet conclude	Needed to allocate
P7	₹20,000	Claim for S102; UTR unreadable; no credit found	Claim: Awaiting bank	None	Whether any money was paid. May be P2	Readable UTR matching a credit. Not found after 3 working days → Ops
P8	−₹5,000	Bank debit referencing P3 (FR-20)	P3 partly reversed	S103 admission + ₹5,000 · credit note · P3 receipt marked	Reason: refund, chargeback or bank error	Finance classifies; student notified
P9	₹25,000	Bank-confirmed; no ID; payer name = S104 full name	Suggested (Medium)	None	Owner. A name is not identity	Student ID from payer, or payer account confirmed by S104. Finance approves

Summary	Records	Amount
Bank-confirmed credits	P1, P2, P3, P6, P9	₹1,00,000 (₹95,000 net of P8)
Posted automatically	P1, P3 (+ P8 reversal)	₹25,000 credit, −₹5,000 reversal
Awaiting human decision	P2, P6, P9	₹75,000
Not money / not verified	P4, P5, P7	₹40,000. All three would have been credited under the APM draft

Controls that prevent incorrect balances and duplicate receipts: FR-02 (duplicates), BR1–BR2 (evidence), FR-23 (append-only ledger), FR-24 (one receipt per allocation), BR8 (maker-checker), FR-10 (overpayment hold), FR-28 (daily tie-out).

10. Data requirements

Minimum fields the product needs. Field names are indicative; engineering owns the schema.

Entity	Fields
Payment	Payment ID · source · UTR / transaction ID · credit or debit · amount · value date · payer name · payer account / VPA (masked) · payer contact (if present) · reference / narration · original payment (for debits) · import batch · status · created at
Allocation	Allocation ID · payment · student · fee component · amount · method (auto / manual) · evidence type and confidence · reason · proposed by · approved by · status · timestamps
Ledger entry	Entry ID · student · fee component · type (charge / payment / reversal / refund / adjustment) · amount · source (allocation or SMS charge) · posted at · posted by
Fee component	Student · type (admission, tuition instalment n) · amount due · due date. Source: SMS
Student	Student ID · name · status · registered contact · guardian contact · email. Source: SMS
Verified payer	Payer account / VPA · linked student(s) · verified by · verification method · date
Claim	Claim ID · student · submitted by · UTR · amount · payment date · payer name · screenshot · status · linked payment · recheck count
Cash note	Note ID · student · amount · collected by · collected at · acknowledgement-slip number · deposit reference · linked payment · verified by · status

Entity	Fields
Receipt / credit note	Number (sequential) · type · allocation · student · amount · issued at · status (valid / partly reversed / cancelled) · replaced by
Mismatch ticket	Ticket ID · student · ledger line or claim · description · assignee · status · due
Audit record	Entity and ID · action · actor (user or system) · timestamp · before · after · reason

11. Permissions and approvals

Capability	Student	Counsellor	Ops exec	Ops lead	Finance	Fin. head	Admin
View ledger and receipts	Own	Assigned	All	All	All	All	All
Submit claim	✓	✓	✓	✓	–	–	–
Create cash note	–	✓	–	–	–	–	–
Propose allocation / split	–	–	✓	✓	✓	–	–
Approve: High confidence, ≤ ₹25,000	–	–	–	✓	✓	✓	–
Approve: Medium / Low, split, > ₹25,000, reallocation	–	–	–	–	✓	✓	–
Verify cash	–	–	–	–	✓	✓	–
Propose refund / approve refund	–	–	–	–	Propose	Approve	–
View unmasked payer account	–	–	–	–	✓	✓	–
Change parameters / roles	–	–	–	–	–	Approve	Propose
Export audit log	–	–	–	✓	✓	✓	✓
Edit balance or ledger entry	**Not available to any role**						

Permissions are enforced on the server, not only hidden in the interface. BR8 applies to every approval.

12. Screens

Low-fidelity wireframes, populated with the sample data. Numbered annotations on each image explain behaviour.

12.1 Unmatched Payment Dashboard (Ops, Finance)

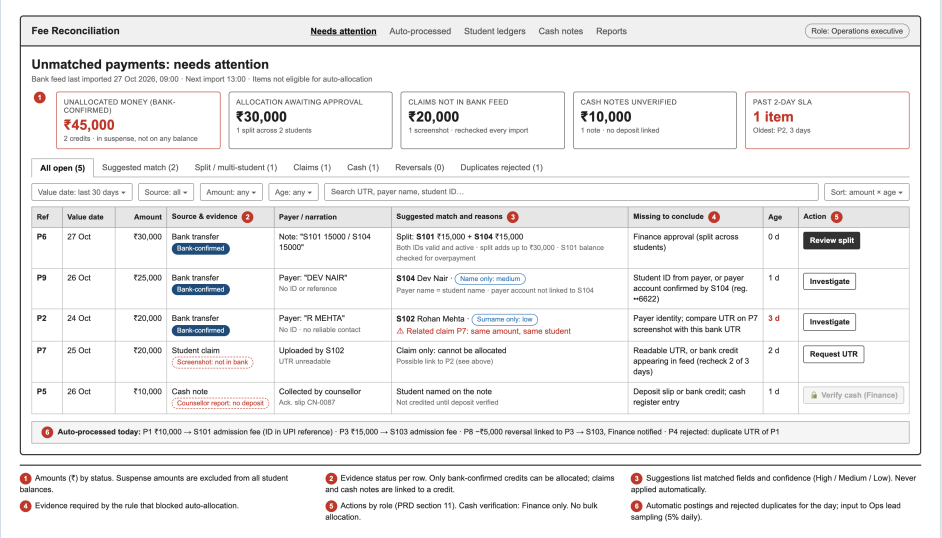


Figure 2. Suspense queue with P6, P9, P2, P7, P5. Bottom strip: today's automatic postings for sampling.

Element	Requirement
Summary tiles	Amounts (₹) for unidentified, pending approval, claims awaiting bank, unverified cash; count past SLA
Queue	Columns: reference, value date, amount, source and evidence status, payer / narration, suggestion with reasons and confidence, missing evidence, age, action. Default sort: amount × age
Filters	Status tab, date range, source, amount band, age; search by UTR, payer name, student ID
Actions	Role-dependent (section 11). No bulk allocation
States	Empty queue; feed stale banner (last import time); item locked while another user has it open

12.2 Student Fee Ledger (Student, staff)

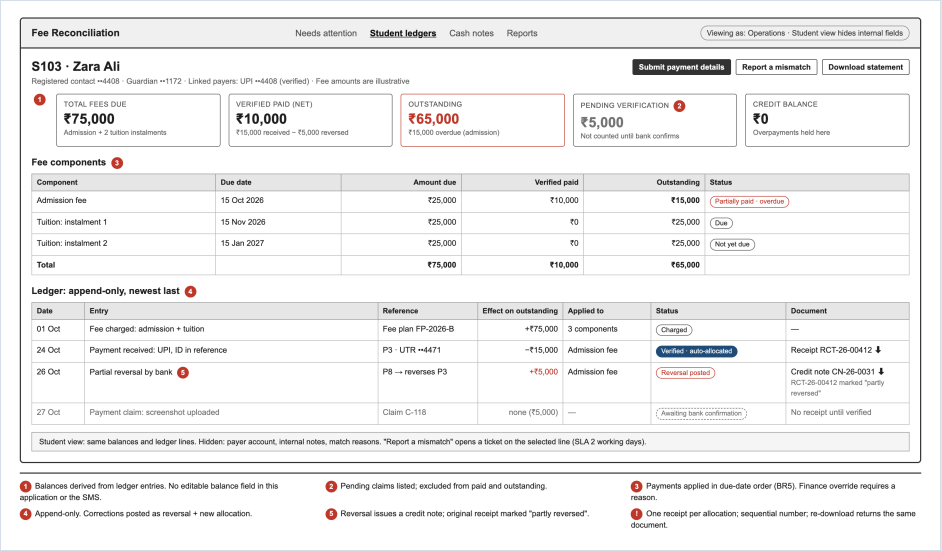


Figure 3. S103: P3 allocated, P8 partial reversal, one pending claim. Fee amounts are illustrative.

Element	Requirement
Summary	Total due, verified paid (net), outstanding (overdue part shown), pending verification, credit balance
Fee components	Component, due date, amount due, verified paid, outstanding, status
Ledger	Date, entry type, reference, effect on outstanding, component, status, document link. Pending claims are listed but excluded from totals
Student view	Hides payer account, internal notes and match reasons. Actions: submit payment details, report a mismatch, download statement

12.3 Payment Reconciliation Detail (Ops, Finance)

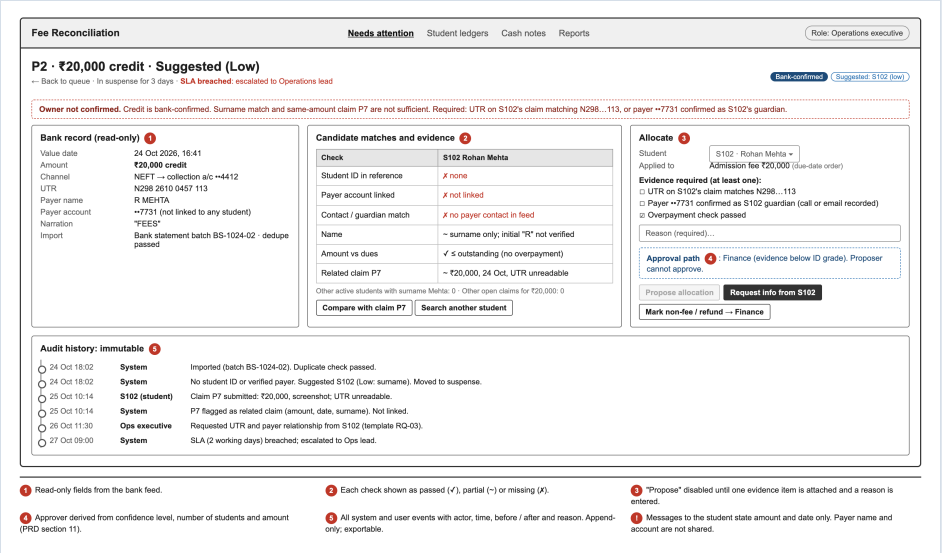


Figure 4. P2 under investigation, with related claim P7.

Element	Requirement
Status banner	States what is confirmed (money received) and what is not (owner), and the evidence needed
Bank record	Read-only fields from the feed
Candidates	Per student: each check shown as passed / partial / missing; related claims; other students with the same surname
Allocate	Student, component split, evidence checklist (one item required for Medium / Low), reason (required). Shows the approval path. "Propose" disabled until complete
Other actions	Request information (template); mark non-fee / route to Finance for refund
Audit history	All system and user events for this payment, newest last

13. Errors and exceptions

Condition	Handling	User message / indicator
Feed import fails or is incomplete	Nothing from the batch is posted; retried; Finance alerted	Banner: "Bank data last updated [time]"
Feed older than 24 hours	Claim Not-found transitions paused	Same banner
Malformed row	Quarantined with reason (FR-04)	Listed in import report
Student ID in reference not found	Unidentified, tagged "unknown ID"; re-matched on student creation	Queue: "ID not found"
Student inactive or withdrawn	Not auto-allocated; routed to Finance	Queue: "Inactive student"
Split total ≠ credit amount	Proposal blocked	"Lines total ₹X; payment is ₹Y"
Amount exceeds outstanding	Excess to credit balance; Finance task	Warning before proposing
Concurrent approval of the same item	First approval applies; second rejected	"Already allocated by [user] at [time]"
Proposer attempts approval	Blocked (BR8)	"Needs a different approver"
Receipt generation fails	Retried; never assigns a second number to the same allocation	Ledger: "Receipt being generated"

Condition	Handling	User message / indicator
SMS sync fails	Retried; alert after 3 failures	Student record: "Sync pending"
Notification fails	Retried on another channel; logged	Audit log only
Invalid screenshot	Rejected if not JPG / PNG / PDF or > 5 MB	Inline error; other fields kept
UTR claimed twice / already allocated	Conflict flag to Ops (FR-16)	"This payment is already claimed. Ops will contact you"
Debit without reference / exceeding allocation	Finance queue, not posted (FR-21)	Finance task
Unauthorised action	Rejected on server; logged	"You do not have permission for this action"

14. Non-functional requirements

ID	Area	Requirement
NFR-01	Accuracy	Re-importing any file or feed window produces no new ledger entries or receipts
NFR-02	Timeliness	Auto-allocated credits are posted and receipted within 15 minutes of import
NFR-03	Volume	Handles peak admission-season volume (figure to be supplied by Finance) without slowing the queue screens
NFR-04	Audit retention	Ledger, receipts and audit records retained for the statutory period for books of account (Finance to confirm; at least 8 years)
NFR-05	Privacy	Payer details masked per BR11; access to unmasked data logged; screenshots visible only to the student, Ops and Finance
NFR-06	Security	Staff sign in through the institution's login; role changes logged and approved
NFR-07	Availability	Student ledger available during admission season; a feed outage does not block viewing or claims
NFR-08	Monitoring	Alerts for feed failure, sync failure, and a tie-out difference not equal to ₹0

15. Acceptance criteria and test scenarios

15.1 Acceptance criteria

ID	Given	When	Then
AC1	A bank credit referencing one active student ID, amount ≤ outstanding	It is imported	Allocated, posted and receipted within 15 min; no human step; included in sample list
AC2	A transaction with an existing source + UTR	It is imported	No ledger entry, no receipt; logged as duplicate
AC3	A claim or cash note with no linked bank credit	At any time	Paid and outstanding amounts unchanged; no receipt exists
AC4	A claim whose UTR and amount match an unallocated credit	The claim is submitted or the credit is imported	Linked; allocation proposed to the claiming student; Ops lead approval required
AC5	An unlinked claim	3 working days pass with fresh imports	Status Not found; assigned to Ops
AC6	A Medium or Low suggestion	Ops proposes it	Cannot post without Finance approval
AC7	A split proposal	Line total ≠ credit amount	Proposal blocked with the difference shown

ID	Given	When	Then
AC8	A posted credit	A bank debit referencing it is imported	Reversal entry on the same student and component; credit note; original receipt marked; Finance flag
AC9	A user proposed an allocation	The same user attempts approval	Rejected by the server; event logged
AC10	Any student	Balance is displayed or synced	Equals the sum of that student's ledger entries; no UI or API can set it
AC11	Any create, approve, reverse or parameter change	It completes	Audit record with actor, time, before / after, reason; cannot be edited or deleted
AC12	A credit exceeding outstanding	It is allocated	Excess held as credit balance; Finance task created
AC13	A credit referencing an unregistered student ID	The student record is created	Matching re-runs and the credit is allocated per BR3
AC14	An allocated receipt	Downloaded again	Same number and document

15.2 Test scenarios

ID	Type	Scenario	Expected result
T-SAMPLE	Fixture	Load P1–P9 and S101–S104 with test fee components	Results match section 9 exactly
T1	Normal	Import P1	S101 admission −₹10,000; receipt; notification
T2	Normal	Ops proposes P6 split; Finance approves	Two allocations, two receipts; overpayment check on S101
T3	Normal	Student submits claim with P9's UTR as S104	Linked; High confidence; Ops lead approves; receipt
T4	Exception	Import P4; then re-import the full statement	No new entries; duplicates logged
T5	Exception	P7 with no matching credit for 3 working days	No balance change; Not found; assigned to Ops
T6	Exception	Counsellor attempts to mark P5 paid	Not possible; only a cash note; Finance verifies against deposit
T7	Exception	P8 after P3 receipt issued	Reversal; credit note; S103 outstanding +₹5,000
T8	Exception	Ops executive proposes P2 → S102 and approves it	Approval rejected; Finance required
T9	Exception	Two approvers act on the same item at once	One allocation; second user sees conflict
T10	Exception	Reallocate a posted payment to the correct student	Reversal + new allocation; old receipt cancelled; new issued; both notified
T11	Exception	Feed down for 24 hours	Banner shown; claims not moved to Not found

16. Release plan and open items

16.1 Rollout

1. Migrate opening balances; Finance ties them out to the previous term's books.
2. Parallel run for 2 weeks: system and manual process side by side; daily comparison; every difference explained.

3. Cutover: SMS balance fields set to read-only; students given the ledger link and payment-reference instructions.
4. Weeks 1–4 after cutover: daily review of sample list, suspense ageing and tickets; parameters tuned with Finance.

16.2 Release checklist

Area	Criterion
Functional	All High-priority requirements built; AC1–AC14 and T-SAMPLE, T1–T11 pass in staging
Finance	Rules and parameters signed off; opening balances tied out; parallel run shows no unexplained difference
Controls	Permission tests pass for every role; audit records verified tamper-proof
Operations	Ops and Finance trained; notification templates approved; runbook and support owner assigned
Fallback	Rollback plan agreed: manual process and SMS balance editing can be restored within one working day

16.3 Questions for engineering

#	Question	Needed for
Q1	Which feeds are available (bank API, statement file, gateway / UPI webhooks), at what frequency, and which fields are reliable?	FR-01, FR-05, AS1
Q2	Can the SMS accept balance updates by API and lock manual balance edits?	D6, FR-26
Q3	How will a payment that appears in both the gateway feed and the bank statement be recognised as one transaction?	FR-02
Q4	Can fee components and due dates be read per student from the SMS?	FR-09
Q5	How will the ledger and audit log be protected from edits, and receipt numbers stay sequential under concurrent approvals?	FR-23, FR-24, FR-29

16.4 Open decisions

Decision	Owner	Due
Pursue per-student virtual accounts / payment links with bank or gateway (phase 2)	Founder	6 Nov 2026
Suspense credits unresolved at 30 days are refunded to the source account	Founder	6 Nov 2026
Parameter defaults in section 7.3	Finance head	29 Oct 2026
System of record for balances (D6)	PM + Engineering lead	Walkthrough

Appendix A. Gap analysis of the APM draft

Category	Gap	Resolved in	Blocks build
Business rules	"Paid" not defined; screenshots and counsellor entries treated as payment	D1, BR1–BR2	**Yes**
Business rules	No fee component order, overpayment or credit-balance rule	BR5–BR6	**Yes**
User journeys	Only student upload defined; no Ops, Finance or approval journey	Section 8	**Yes**

Category	Gap	Resolved in	Blocks build
User journeys	Student cannot see pending vs verified or raise a mismatch	12.2, FR-30	No
Technical / integration	No bank / gateway feed; no source of truth for receipt of money	FR-01, Q1	**Yes**
Technical / integration	System of record and sync with SMS undefined	D6, FR-26, Q2	**Yes**
Data accuracy / control	Name fallback allocates to the wrong student	D3, 7.2	**Yes**
Data accuracy / control	Counsellors can edit payments and mark cash paid; no maker-checker or audit	D5, BR8, FR-29	**Yes**
Exception handling	No duplicate detection	FR-02–FR-03	**Yes**
Exception handling	Splits, partial payments, reversals, refunds, early payments not handled	FR-08, FR-20–FR-22, BR9	**Yes**
Usability / reporting	Dashboard lacks UTR, evidence status, component, age; no ageing or tie-out reports	12.1, FR-28	No
Usability / reporting	Goal "eliminate manual reconciliation" and 2-week estimate not measurable or validated	Section 2	**Yes**

Appendix B. APM feedback

Item	Feedback
Strengths	Correct problem selection. Covered cash and manual fallback. Included student self-service and receipts.
Corrections (top 5)	1. Screenshot is evidence, not payment; bank feed is the source of truth. 2. No allocation on name. 3. Separate recording from approval; add audit. 4. Define duplicates, splits, partial payments, reversals, refunds. 5. Replace "eliminate reconciliation in 2 weeks" with metrics; estimate comes from engineering.
Returned to APM	Write T-SAMPLE expected results independently and compare with section 9 · shadow Ops for one day and collect baselines (section 2.3) · draft notification and request-for-information templates · validate section 10 fields against SMS data
PM-owned decisions	Evidence standard and approval matrix (with Finance) · phase 1 scope · system-of-record proposal (with engineering)
Pre-engineering check	Finance sign-off on section 7 · each AC testable · section 9 reproduced by a second person · every open item has owner and date
Development plan	Write a failure-mode note for each rule before review · weekly 30-min spec review against a checklist (evidence, exceptions, permissions, audit, metrics) · APM leads the next engineering walkthrough with PM observing

Appendix C. Founder update format

Sent Monday and Thursday. Example: Monday, week of 26 Oct 2026.

Section	Content
Status	On track for engineering kickoff on 3 Nov
Completed	APM draft review (12 gaps, 10 blocking) · PRD v1.0 · 3 wireframes · sample data walkthrough

Section	Content
Pending requirements	Finance sign-off on parameters (Wed) · feed field list from bank (Thu) · notification templates (APM, Fri)
Decisions made	Bank feed is sole proof of payment · no allocation on name · counsellors record cash, Finance verifies · balances derived from ledger
Risks / dependencies	Bank API lead time unknown (fallback: statement file) · SMS API may need vendor work · opening balance migration
Testing / readiness	14 AC and 12 test scenarios written · 2-week parallel run planned · release checklist agreed
Founder decisions	1. Pursue per-student virtual accounts / payment links (phase 2), by 6 Nov. 2. Approve refund-to-source for suspense credits at 30 days, by 6 Nov.

Not escalated (resolved by PM / Finance): parameter values, screen design, test plan, APM task allocation.